

Extract, then decide: a modular pipeline of LLMs and rules to identify seizure frequency in epilepsy clinic letters

A dissertation submitted in partial fulfilment of
The requirement for the degree of
MASTER OF SCIENCE in Artificial Intelligence
in
The Queen's University of Belfast
by
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4th September 2026

Declaration of Academic Integrity

I certify that the submission is my own work, all sources are correctly attributed, and the contribution of any AI technologies is fully acknowledged.

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Name : CONOR BROWN

Student Number: 40466218

A handwritten signature in black ink. The word "Brown" is written in a bold, slightly cursive font. It is enclosed within a large, sweeping, horizontal loop that starts above the first letter, goes under the entire word, and ends above the last letter.

Student's signature:

Date of submission: 4th September 2026

Extract, then decide: a modular pipeline of LLMs and rules to identify seizure frequency in epilepsy clinic letters

Abstract—Seizure frequency patterns are recorded in clinical notes stored in electronic health records (EHRs). Manual chart review is required to extract this information for research, but this is expensive and error-prone. Rules-based systems struggle with the lexical variety of seizure frequency descriptions. Large language models (LLMs) are substantially better at seizure frequency extraction, but fine-tuning a model for this task requires substantial data and computation. This paper proposes a training-free pipeline with two stages. One LLM call **extracts** every candidate seizure frequency event in the letter into a structured record, each with its exact evidence span, a category and a canonical label, plus a provisional answer. A second stage then **decides** the final answer from that record by applying a fixed policy. The decision stage was performed in two ways on the same extraction record: by recorded rules (**Hybrid**) and by a second LLM call (**LLM-only**). Evaluation used 1,200 synthetic epilepsy letters and was compared with the benchmark previously reported for a fine-tuned LLM on a different held-out sample of the same corpus. Both **Hybrid** (Purist F1: 0.86) and **LLM-only** (0.85) exceeded the reported benchmark (0.81) without any training or fine-tuning, and both improved on the extraction call’s own provisional answer (0.79). Ablations of the extraction prompt show that its examples, closed label forms and evidence obligation each contribute to that result. On this synthetic benchmark, prompted extraction followed by a recorded decision policy is a practical alternative to fine-tuning.

Index Terms—information extraction, hybrid pipeline, large language models, seizure frequency, epilepsy

I. INTRODUCTION

Epilepsy affects c. 1% of the UK population [1] and is characterised by recurrent, unprovoked seizures [2]. Seizure frequency is a key indicator of the patient’s disease burden and is commonly used in clinical care to determine the appropriate treatment and as a primary outcome in clinical trials. Despite the critical role of this information, the only place it is routinely recorded is in clinic letters. Clinic letters are unstructured narrative reports of a patient’s recent visit to the epilepsy clinic, and they often combine descriptions of past diagnoses, current symptoms and future treatment. This information can be used to conduct retrospective studies or to identify patient cohorts for clinical trials, but extracting it into that structured form requires experts to conduct costly, error-prone manual chart review.

Natural language processing (NLP) has been used to automatically extract key information from clinic letters. Rules-based methods successfully extracted structured facts about medications and diagnoses, but struggled with the imprecise language and diverse forms of seizure frequency descriptions

[3]. *2 to 3 seizures a week*, *frequent bouts of seizures* and *4 since her last visit* are all valid descriptions of the same fact, but designing rules to accommodate these completely and consistently is challenging and time-consuming. Rules-based systems are highly valued by clinicians because they produce interpretable results and can flexibly incorporate local domain expertise, but a different kind of tool is needed for these semantically complex tasks.

Large language models (LLMs) perform substantially better on semantically complex tasks from instructions and a few examples alone [11], [12], including seizure frequency extraction [4]. However, unlike rules-based methods, the outputs of LLMs are not constrained to the exact text they were given. This presents new challenges for the task of information extraction (IE). Past work has shown that LLMs can invent evidence even when given the specific task of estimating a patient’s seizure frequency from the letter. In one case an LLM provided diagnostic advice on prescriptions instead of citing the seizure frequency [5]. This phenomenon of ‘hallucination’ has been documented as a common error mode for LLMs and is a significant risk when applied to clinical information at scale. Fine-tuning an LLM on the task minimises this error but requires substantial training data and computation.

Prior work has utilised rules as a light post-processing layer to repair malformed outputs from LLMs [5], but none examine a hybrid method of LLM and rules as an alternative to fine-tuning LLMs. Rules are very effective at processing information with transparent, configurable logic. LLMs are very effective at extracting rich information from semantically dense sources. Leveraging both capabilities at distinct stages can combat the concerns about LLMs being a black box that produces hallucinations, while providing greater semantic depth with far less time spent developing rules. A hybrid method can produce a more transparent, flexible system.

Extracting a single label is not a single step. A clinic letter can describe several seizure events with varying precision and on different occasions, so the system must first **extract** every candidate fact with its evidence, then **decide** which candidate, or which combination of them, answers the question. Which to keep is a clinical policy that varies between users. Keeping the two stages apart captures more evidence than a single answer would, lets one extraction serve several policies, and makes the final answer explainable: a reader sees the candidates, the policy and the label it produced.

The primary objective of this paper is to evaluate whether a

training-free pipeline of a prompted LLM and recorded rules can extract seizure frequency patterns from epilepsy letters as accurately as the fine-tuned LLM previously reported on the same corpus. The secondary objective is to test the extract-then-decide design itself: whether a decision stage applied to the extraction record improves on the extraction call’s own provisional answer, whether it matters if rules or a second LLM call apply the policy, and which ingredients of the extraction prompt the result depends on. If both objectives are met, the paper contributes a new method for seizure frequency extraction and shows where each component helps, to support future research on a more general form of the method.

To answer these questions, the pipeline was scored after each stage on 1,200 synthetic epilepsy clinic letters from Gan *et al.* [3], each with a single expert-annotated label for the patient’s current seizure frequency pattern (Section III-A). Results are compared with the benchmark Gan *et al.* report for their fine-tuned LLM. The two papers use different held-out samples from the same synthetic corpus, so this is a bounded comparison with a previously reported benchmark rather than a paired evaluation on identical letters; Section IV-A reports the uncertainty this introduces.

II. LITERATURE REVIEW

The best approach to extract seizure frequency information from epilepsy clinic letters is an active area of research, with experiments using rules-based methods [4], [6], transformers-based methods [3], [5], [7], [8] and a combination of both [9], [10] all published this decade. Rules-based methods struggled with the lexical variety of seizure frequency patterns. A key challenge with both rules-based methods and BERT-based methods was their failure to generalise to letters from a different hospital. [6], [10]. LLMs were the first method to demonstrate strong generalisation to a new dataset [3], but this was achieved by fine-tuning a model on over 1,000 synthetic letters. This paper explores whether similar performance can be achieved without any additional training or fine-tuning.

Rules were used to normalise frequencies and dates as a separate layer in two BERT-based pipelines [9], [10], and as minimal repair of malformed LLM output [5], but no study has given rules a decision role after a modern LLM. This paper explores whether LLM instructions and examples combined with recorded decision rules can replace the need for fine-tuning.

Clinical IE has been defined as the automatic extraction and encoding of clinical information from text [13]: recognising a mention and normalising it into a standard form [14]. Modern LLMs do both in one step when asked for a normalised label such as *4 per month* from a raw statement such as *about once a week*. A further step exists in many clinical extraction pipelines but goes unmentioned: deciding between candidates. In a clinic letter with several seizure frequency statements, producing a single label for the patient’s current status requires the system both to extract the relevant statements and to decide which one best answers the question. Prior rules-based systems made this a distinct stage [15], and

Thank you for asking me to review your patient. I am concerned that her epilepsy has worsened in recent months, with a clear pattern of deterioration. She and her partner report increasing clusters of events correlating with the perimenstrual period, suggesting a catamenial component. In particular, she describes brief focal episodes with preserved awareness that tend to occur as she is settling to sleep, at the point of drifting off, now happening on most nights of the week. These events are accompanied by a rising epigastric sensation, facial pallor, and right-hand automatisms lasting 30–60 seconds, followed by fatigue and headache. In addition, she has had three generalised tonic–clonic seizures in the last six weeks, two of which occurred during the same perimenstrual window.

Fig. 1: An example clinic letter with two distinct seizure frequency patterns highlighted as potential candidates.

clinical concept normalisation routinely separates a high-recall candidate generation stage from a precise ranking stage that chooses among the candidates [21]. LLM pipelines instead make the decision inside the extraction prompt. This paper explores whether separating *extract* from *decide* in a modular pipeline can deliver comparable performance while preserving more of the evidence for downstream use.

Academic papers tend to focus exclusively on benchmark accuracy while practitioners tend to value transparency, configurability and efficiency at the cost of some accuracy [15], [16]. Model interpretability is also a critical factor clinical settings [14]. Clinicians are trained to use deductive logic to draw conclusions and they expect systems to expose transparent, interpretable evidence that they can reliably use to make decisions that demand high accountability. Past work utilises evidence spans as a training input [3] for fine-tuning or as data exhaust as part of using a rule-based method [4]. This paper explores whether preserving a transparent evidence chain from exact text to final answer can make the output both more interpretable and more accurate.

III. METHODS

A. Data

The dataset is a collection of epilepsy clinic letters which record a UK patient’s consultation with an epilepsy specialist, often a neurologist or specialist nurse [3] The NHS requires that clinic letters are sent to the patient’s GP after all outpatient visits such as these. They are often the only record of a patient’s diagnoses, treatments and outcomes in neurological disorders, and they are written in narrative form. An example clinic letter is shown in Figure 1.

Gan *et al.* [3] produced this dataset of synthetic letters modelled on real clinic letters from King’s College London hospital. The real letters were used to create 10 template letters and they were combined with 500 short descriptions of seizure frequency patterns produced by GPT-5 to create the full set of 15,099 letters. A sample of 1,200 letters of c. 400 words was used in this paper. Trained experts annotated each letter with

TABLE I: Dataset summary

		Development	Test
Number of letters		750	450
Seizure frequency	Frequent	51%	50%
	Infrequent	17%	18%
	Unknown	17%	17%
	Seizure free	15%	15%

a single gold-standard label representing the patient’s current seizure frequency status, and with one supporting quote.

Minimal preprocessing was applied: letters were cleaned only to parse quotes, new lines, tabs, the $\leq$ sign and nulls, and were ingested in full with no tokenisation, section-splitting or truncation.

The dataset was split into a development set of 750 letters and a test set of 450 letters, stratified by their seizure frequency category (frequent, infrequent, unknown, seizure free) to balance the distribution across both splits. The exact breakdown can be seen in Table I.

B. Architecture

The pipeline has two stages: *extract*, then *decide*. *Extract* is one LLM call that reads the full letter and returns a structured record of every candidate seizure frequency event, each with its exact quoted evidence span, a category and a label in one of the canonical forms used by the gold standard. The same call also proposes a provisional answer from its own candidates. This is what a single-prompt system would return, so scoring it gives a one-prompt baseline; the *decide* stage then confirms or replaces it. *Decide* receives that record, not the letter, and produces the final answer by applying a fixed decision policy, either picking one candidate or combining several. Figure 2 illustrates the pipeline with a development-set example.

The decision stage was performed in two ways on the same saved extraction record (Table II). In *Hybrid*, recorded rules apply the policy. In *LLM-only*, a second LLM call applies the same policy, expressed as written instructions and worked examples. The only difference between the two is therefore who performs *decide*; the extraction call, its output and the policy are shared. Because the decision stage works from the saved record, both executors were replayed on the same extraction output without a new model call, so any difference between them is attributable to the decision stage alone.

The pipeline is a fixed sequence of calls and functions. No LLM call uses tools, chooses the next step or invokes another model, so the system is not described as agentic; each LLM call is a prompt with a fixed input and a schema-validated JSON output. A rules-only baseline and several further decompositions of the LLM work were also built and are described in the supporting materials; they are not part of the main comparison.

C. LLM Prompt Design

The most consequential component of the pipeline is the *extract* prompt used for the first call. It has four ingredients,

TABLE II: Interface contract between the two stages

	Extract	Decide
Input	Full letter	Candidate record and decision policy; not the letter
Output	Candidate record: events, each with quoted span, category and canonical label; provisional answer	Final label and its supporting event(s)
Performed by	LLM call 1	Hybrid: recorded rules. LLM-only: LLM call 2

Hybrid and LLM-only share the same first call and the same saved record. Full prompts and schemas are in the supporting materials.

TABLE III: Decision policy

When	Do
Several current seizure types are present	Choose the highest current or recent frequency.
The letter gives an overall current count and a breakdown by type	Choose the overall count.
A seizure-free statement sits with other current seizure-like events	Keep seizure-free separate from unknown or last-event statements.

each of which is removed in turn in Section IV-C:

- 1) *Instructions* to identify relevant seizure frequency events and to propose a provisional answer.
- 2) *Examples* of how events should be represented.
- 3) *Allowed labels*: a closed list of the canonical label forms used by the gold standard.
- 4) An *evidence obligation*: schema fields for the evidence span of every event, with the instruction that each span must be an exact quote from the letter.

The output schema represents multiple seizure frequency patterns with exact evidence, so that rules or a second LLM call can work from the candidates rather than from the letter. The same decision policy (Table III) is applied twice: provisionally by the model inside the extraction prompt, and finally at *decide*. For example, if the model finds *3 seizures in March, 6 in May* and provisionally answers with the most recent count, the *decide* stage can replace that with the total over the whole period, *9 per 3 months*, using the evidence the model already found. The second-call *decide* prompt states the same policy as written instructions with worked examples and receives only the candidate record.

The model is instructed to assign each event to one of six categories (a countable or range *frequency rate*, a *cluster frequency*, a *seizure-free* statement, a dated *last event* without a rate, an *unknown frequency* where seizures are discussed without a usable rate, and *no reference*) and to write its label in one of the canonical forms listed in the prompt. The category definitions, label forms and full prompt are in the supporting materials.

D. Rule Design

In *Hybrid*, recorded rules perform *decide*. Each rule is a named function with *regex* matching and if-then logic that reads the candidate record the model returned, never the letter.

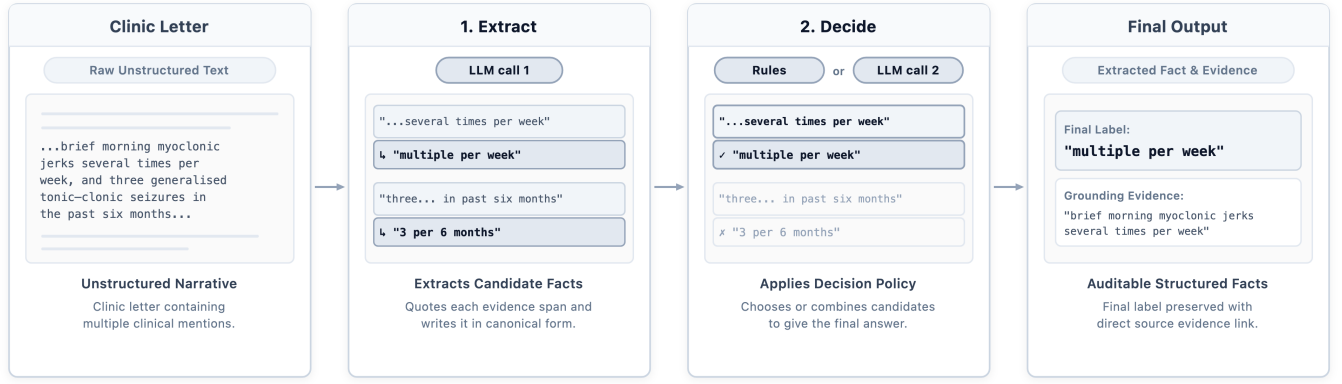


Fig. 2: A clinic letter passes through two stages (extract, then decide) to produce a structured label with grounding evidence. Extract is one LLM call. Decide works from the resulting record and is performed by recorded rules (Hybrid) or by a second LLM call (LLM-only).

Rules first rewrite any label the model left in a non-canonical form (a step that changes the score by at most 0.01 and is detailed in the supporting materials), then apply the decision policy in Table III. Three kinds of decision rule are used. A *gate* rejects a provisional answer that the policy forbids: if the model finds both *seizure free for eight months* and *two seizures in the past fortnight* but provisionally answers seizure-free, the gate rejects seizure-free while other current seizure-like events remain. A *reselect* rule chooses a different already-extracted event, such as a frequency candidate when the model chose unknown. A *rewrite* rule derives a new label from the extracted events: for seizure diaries, date and numeric logic totals the events over the stated period, so *May x0, Jun x0, Jul x1, Aug x0, Sep x1, Oct x2* becomes *4 per 6 months*. Every rule is recorded with the event it acted on, so a final label can be traced to the model’s quoted evidence and the rule that decided it. Rules do not scan the letter for new candidates; every rule acts on the events the model extracted.

E. Pipeline Optimisation

The development set was used to optimise the two key elements of the pipeline: the LLM prompt instructions and the rules. The focus of the optimisation process was to improve accuracy while keeping the instructions and rules simple and generalisable. Qualitative error analysis was conducted to diagnose the cause of poor performance and identify common error modes. Hypotheses were developed and tested to tune the prompt and rules iteratively. Improving a rule or instruction for one class of error often harmed another, because the annotation policies interact. Separating *extract* from *decide* kept the policy on which facts to extract apart from the policy on which to choose, and made it possible to score the pipeline after each stage and locate bottlenecks. The supporting materials give a fuller account of this process.

F. Evaluation

Every paper reporting seizure frequency extraction this decade used F1 as its primary metric, but they differ in

what counts as a correct frequency label. It is not obvious whether *2 per month*, *3 per month* and *once a week* should all be accepted when the gold-standard label is *multiple per month*. To allow comparison with Gan *et al.* [3], this paper adopts their two evaluation settings: a fine-grained *purist* setting and a coarser *pragmatic* setting. If the gold label is *twice a week* and the prediction is *once a week*, that is a correct *pragmatic* answer, since both fall in the frequent category, but an incorrect *purist* answer. Table V lists the categories in each setting.

Micro-F1 is reported for both settings, because the categories are badly imbalanced (some have over 100 letters, some fewer than 10; Table V) and micro-F1 gives each letter equal weight. Each letter has exactly one gold label and one prediction, so micro-F1 here equals the share of letters answered correctly. The more precise *purist* F1 is the primary metric.

The pipeline is evaluated on a held-out test set of 450 letters that were never used to tune prompts, rules or any other aspect of the pipeline. Results are compared against the evaluation set labelled Synthetic (1,166) in Gan *et al.* [3], a different held-out sample from the same synthetic corpus, distinct from their fine-tuning data.

The primary model is Gemini 3.7 Flash. It is the only model used for the second decision call (LLM-only), for the prompt ablations and for the temperature and thinking-level ablations. Six models were evaluated as the extraction call in Hybrid (Section IV-D). Elsewhere, Hybrid results are from Gemini at temperature 0 with low thinking effort. The held-out results are reported as aggregate totals only; individual test letters and their outputs were not inspected.

IV. RESULTS

A. Benchmark Performance

The primary objective of this paper is to evaluate whether a training-free pipeline of a prompted LLM and rules can match a fine-tuned LLM at extracting seizure frequency patterns

TABLE IV: Comparison with previously reported results

Author	Year	Method	Purist F1	Pragmatic
Holgate et al.	2024	Prompted LLM	-	0.73
Holgate et al.	2025	Fine-tuned LLM	0.68	0.81
Gan et al.	2026	Fine-tuned LLM	0.81	0.85
This paper	2026	Hybrid (LLM + rules)	0.86	0.88

Each row was evaluated on a different held-out sample; only Gan et al. and this paper share a corpus and metric definitions.

TABLE V: Hybrid: Purist and Pragmatic categories (test set, $n = 450$)

Purist	n	Purist F1	Pragmatic F1
Daily	41	0.92	Frequent: 0.95
More than weekly, less than daily	123	0.91	
Once a week	5	0.89	
More than monthly, less than weekly	58	0.90	
Once a month	18	0.94	Infrequent: 0.82
More than every 6 months, less than monthly	55	0.81	
Once every 6 months	1	0.00	
Less than once every 6 months	6	0.60	
Unknown	76	0.74	Unknown: 0.74
Seizure free	67	0.90	Seizure free: 0.90

Rows with $n < 10$ shown in grey.

from epilepsy letters. Results are compared with the Synthetic (1,166) evaluation reported by Gan *et al.* [3]. The two evaluations use identical metric definitions and letters drawn from the same synthetic corpus, but different held-out samples. The comparison is therefore with a previously reported benchmark, not a paired evaluation on the same letters, and it does not establish a state-of-the-art result.

Hybrid achieves a purist F1 of 0.86 (387/450), above the reported benchmark of 0.81 for the fine-tuned LLM. LLM-only also exceeds it (0.85; 383/450). Because micro-F1 equals per-letter accuracy here, an approximate two-proportion comparison gives a difference of +0.05 for Hybrid (95% CI +0.01 to +0.09, $p = 0.01$) and +0.04 for LLM-only (95% CI 0.00 to +0.08, $p = 0.04$). These intervals assume independent samples from the same distribution and are indicative rather than a controlled comparison. The first LLM call alone, scored on its provisional answer, reaches 0.79, close to the benchmark. Rules applied to its candidates add +0.07 in Hybrid; the second LLM call adds +0.06 in LLM-only.

The pipeline is particularly effective at extracting high-frequency events. Gan *et al.* report *unknown* as their highest scoring category (0.89), with *daily* (0.70) and *more than weekly* (0.68) much lower. In this pipeline, *daily* (0.92) and *more than weekly* (0.91) are among the best performing categories, well above *unknown* (0.74). The full breakdown across purist and pragmatic categories is in Table V.

TABLE VI: Purist F1 after each stage, by decision executor

Decision executor	Provisional answer (after extract)	Final answer (after decide)
Hybrid: rules	0.79 (355)	0.86 (387)
LLM-only: second LLM call	0.79 (355)	0.85 (383)

Correct letters of 450 in parentheses. Both rows share the same extraction call and the same saved record; only *decide* differs.

B. Provisional and Final Answers

Table VI reports purist F1 at two stops: the provisional answer after *extract*, and the final answer after *decide*, for each decision executor. Each stop scores the answer the pipeline would submit if it stopped there. The provisional answer is the extraction call's own proposal, so its score measures the quality of that proposal, not candidate identification alone. The difference between the columns is what *decide* adds to the answer already proposed.

The provisional answer from the single extraction call is 0.79. This call is equivalent to the entire pipeline in most prior LLM studies, which ask the model to extract and decide in one prompt. Both executors then improve on that shared provisional answer at *decide*: rules add +0.07 and the second LLM call adds +0.06. This can mean choosing a different candidate, such as a frequency candidate when the model chose *unknown*, or calculating a new label from the chosen evidence, such as totalling the events in a monthly seizure diary. The two executors differ on 28 of 450 letters (rules correct on 16, the second call on 12); a paired McNemar test gives a difference of +0.01 (95% CI -0.01 to $+0.03$, $p = 0.57$), compatible with no difference. Applying the decision policy to the candidates already extracted is where the pipeline adds value over a single prompt, and on this task it can be performed as well by recorded rules as by a second LLM call. Other decompositions of the LLM work, including a separate label-rewriting call before *decide*, did not improve on this two-stage design and are reported in the supporting materials.

C. Extraction Prompt Ablations

Table VII removes one ingredient of the extraction prompt at a time and scores the Hybrid pipeline at both stops, so the effect of each ingredient can be seen on the provisional answer and after rules have applied the policy. Removing the worked examples, the closed list of allowed label forms, or the evidence obligation each lowers the provisional answer by 8 to 10 letters and the final answer by 17 to 23 letters. The evidence-obligation ablation removes the evidence fields and the exact-quote instruction together as one package; it does not test the quote instruction alone. The decrements are not additive and should not be summed: each row removes one ingredient from the full prompt while the others remain. Rules recover some of each loss but never the whole of it, which shows that the decision stage depends on the quality of the record the extraction call produces.

TABLE VII: Extraction prompt ablations (Hybrid, Gemini, test set)

Extraction prompt	Provisional answer	Final answer
Full prompt	0.79 (355)	0.86 (387)
Without examples	0.77 (345)	0.82 (370)
Without closed allowed-label forms	0.77 (347)	0.81 (364)
Without evidence obligation	0.77 (345)	0.82 (370)

Purist F1 with correct letters of 450 in parentheses. Each row changes only the extraction prompt; the same rules apply the policy at decide. Effects are not additive.

D. Model Comparison

Hybrid shows consistent, incremental gains on each of the six models evaluated as the extraction call (Figure 3). The average gain from the provisional answer to the final answer was +0.07, with Luna showing the largest gain of +0.10. Purist F1 closely tracks general medical capability as measured by Artificial Analysis’ Healthcare and Medical Index. The models cluster into three groups: strong (Gemini 0.86, Grok 0.85), medium (DeepSeek 0.82, Luna 0.79) and weak (Qwen 0.76, Gemma 0.72), which largely reflect model size and cost.

Table VIII reports how well each model kept to the extraction contract. Every model produced a parseable record for at least 442 of 450 letters, and no call failed. The four hosted models quoted an exact substring of the letter as evidence for the selected event in 99% or more of letters; the two local models paraphrased more often (Qwen 97%, Gemma 93%) and needed more non-semantic format repair. An exact span is not proof that the right statement was chosen, but it means that a reviewer can locate the evidence for almost every answer verbatim in the source.

Table IV also lists the two earlier papers that used the same pragmatic metric on different data. Holgate *et al.* [8] ran Llama 2 on local hospital servers with eight Nvidia V100 GPUs and reached a pragmatic F1 of 0.73 without fine-tuning. In Hybrid, Qwen reached a pragmatic F1 of 0.80 running locally on a laptop with a single GPU. This shows that the evidence-backed extraction design can execute on local hardware under the same synthetic task conditions. It is a technical feasibility result, not evidence of performance on real letters or of fitness for deployment.

E. Model Configuration

Raising the reasoning effort level did not improve performance. Gemini’s purist-F1 score was 0.86 at low and 0.84 at high; a paired difference of +0.02, 95% CI -0.01 to 0.04; $p = 0.31$. Extra reasoning adds cost and latency without improving extraction.

The temperature setting also had no significant effect: 0.86 at temperature 0 versus 0.84 at temperature 1 (paired difference +0.02, 95% CI -0.01 to 0.04; $p = 0.20$). Temperature 0 is therefore an appropriate default, and Luna’s fixed temperature of 1 does not invalidate its comparison. The task is tightly

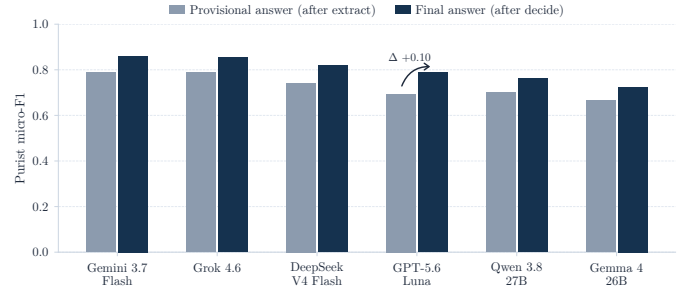


Fig. 3: Hybrid Purist F1 for six models: the LLM’s provisional answer after extract and the final answer after rule-based decide. Every model gains at decide; Luna gains most (0.69 to 0.79).

TABLE VIII: Six models as the extraction call: scores and contract adherence (Hybrid, test set, $n = 450$)

Model	Purist	Prag.	Exact ev.	Repairs	Unparsed
Gemini 3.7 Flash	0.86	0.88	99.8%	0	1
Grok 4.6	0.85	0.89	100%	0	0
DeepSeek V4 Flash	0.82	0.85	99.6%	2	0
GPT-5.6 Luna	0.79	0.82	98.9%	0	1
Qwen 3.8 27B (local)	0.76	0.81	96.7%	3	2
Gemma 4 26B (local)	0.72	0.77	92.9%	15	8

Exact ev.: the quoted span for the selected event is an exact substring of the letter. Repairs: letters needing a non-semantic JSON dialect repair.

Unparsed: letters with no valid record after repair, scored as incorrect. No call failed for any model.

defined by an explicit schema, which limits the expected variability by design and partly explains why neither setting mattered.

F. Error Analysis

The worst performing category was unknown with an F1 score of 0.74. 69% of the errors (37/54) are incorrect answers of unknown. The full error distribution is shown in the confusion matrix (Figure 4).

The three most common errors are infrequent $\rightarrow$ unknown (16), frequent $\rightarrow$ unknown (13), and seizure free $\rightarrow$ unknown (8). The errors reflect a cautious approach when the evidence is ambiguous, for example the model recognised the phrase ‘these episodes crop up most shifts’ as the primary seizure frequency description but refused to make an assumption about how many shifts the person works. The gold-standard label was multiple per week but the model chose unknown. This cautious assessment is justifiable, and because the exact evidence is preserved, downstream users can see the proposed answer and draw their own conclusion from the quoted span.

G. Experimental Environment

Table IX summarises the hardware, software and model settings used. DSPy co-ordinated the model calls and required

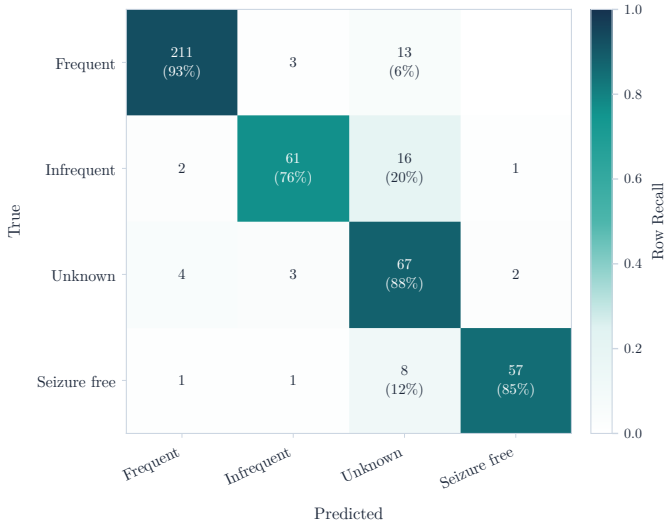


Fig. 4: Hybrid confusion matrix on the pragmatic categories (test set). Most errors are incorrect predictions of unknown.

TABLE IX: Experimental environment

Component	Setting
Local models	Qwen 3.8 27B and Gemma 4 26B served by Ollama 0.32.15 on a Dell XPS 16 9640 (Windows 11, Intel Core Ultra 9 185H, 32 GB RAM, NVIDIA RTX 4070 Laptop GPU, 8 GB VRAM)
Hosted models	Gemini 3.7 Flash, Grok 4.6, GPT-5.6 Luna and DeepSeek V4 Flash, called from a Mac mini M1 (16 GB, macOS 26.5.2) directly (OpenAI, DeepSeek) or via OpenRouter and AI Gateway
Model settings	Temperature 0 (Luna accepts only 1); thinking or effort level low. Gemini ablations: temperature 1; thinking medium and high
Software	Python 3.14.4, DSPy 3.2.1, Pydantic 2.13.4, LiteLLM 1.87.1

structured JSON output, Pydantic validated each record against the schema, and LiteLLM delivered API calls to Ollama and the model routers. Full specifications, API settings and replay artefacts are in the supporting materials.

V. DISCUSSION

This paper shows that, on the Gan synthetic benchmark, a training-free pipeline of one prompted LLM call and a recorded decision policy extracts an epilepsy patient’s current seizure frequency pattern at least as accurately as the fine-tuned LLM previously reported on a different held-out sample of the same corpus [3]. A single prompt designed to extract every candidate with exact evidence, followed by rules or a second call that decide from that evidence, matched that benchmark without any training data or fine-tuning compute. The same design also executed on a laptop-class GPU with open-weight models, at a lower but still competitive score. That is design evidence which later researchers or healthcare organisations can evaluate on their own data and hardware; it is not evidence that the method is ready for local deployment.

The method is particularly effective at identifying the categories that mark the greatest disease burden. F1 scores of 0.92 for daily and 0.91 for more than weekly are around 0.2 higher than the category scores previously reported [3], although on a different sample. Structured seizure frequency of this kind is intended for later research uses: cohort identification for trials, retrospective and longitudinal analyses of treatment response, and outcome variables for predictive models [13], which together can widen the scope of real-world evidence [17]. Those are intended uses, not capabilities demonstrated here; each requires validation on real records first.

Past work has typically tasked the LLM with producing a single answer in a fully normalised form [3]. That discards the semantic depth of the narrative, which clinicians value because it conveys reasoning, uncertainty and vague impressions, and those nuanced observations are more accurate and reliable [18]. Current LLMs can populate a richer record that preserves them: every candidate with its quoted span, the policy applied, and the label it produced. The ablations show that this record is also what the accuracy depends on. A more explainable system is often a more useful system in a clinical setting [14], and here it is not a less accurate one.

There are several limitations. Synthetic data was used for all evaluations, and while past research has shown models fine-tuned on this dataset generalise effectively to real patient data [3], the methods tested here remain untested on real letters; no claim is made about clinical validity, workflow fit or privacy compliance. The benchmark comparison is bounded: the held-out samples differ, so it is not a paired test. Past research has shown that rules tuned on one corpus can generalise poorly to others, so the gains from rules may not transfer to other datasets [6]. The stage scores are cumulative stops rather than isolated stage metrics. Evidence quality was measured only as exact-substring adherence, which does not show that a span semantically supports the label; the gold standard’s single reference quote is a loose comparator for that purpose, since it may be paraphrased or abbreviated. A development-set study of semantic evidence sufficiency, adjudicated from the extracted spans and the policy alone, is specified as the next step before any further held-out evaluation. Finally, the method was applied to one narrow task. The same two stages have been configured for a multi-fact epilepsy inventory task [4], to be reported separately, and future work should evaluate whether the design generalises to other clinical IE tasks and whether a more general framework, in which an LLM chooses how to represent facts and which policy to apply, adds value as reported in oncology [19] and cardiology [20].

VI. AI ETHICAL CONSIDERATIONS

This evaluation used only the synthetic clinic letters released by Gan *et al.* [3]. Those letters are modelled on real UK epilepsy letters but do not contain identifiable patient records. Hospitals prefer processing patient data on their own servers as it is highly sensitive [5]. Two of the six models were run entirely on local hardware to test whether the design can

execute in such a setting. That is a technical feasibility check on synthetic data; it is not a privacy-compliance assessment.

LLMs can invent evidence, so careful consideration was given to the prompt instructions and schema design. The model is required to quote exact evidence spans from the letter, and every later change to the answer is made by a named rule or a recorded second call applied to that quoted evidence. A reviewer can therefore trace any final label back to the text that supports it and to the policy that decided it [14]. This transparency matters more than headline accuracy in a clinical setting, where an unexplained answer cannot be safely acted on.

The letters contain no demographic attributes, so unequal performance across patient groups cannot be tested, and the synthetic corpus may not reflect the language of every clinic. Bias can enter through the models, the prompts and the decision rules; for example, the cautious default to `unknown` (Section IV-F) would systematically under-count patients whose seizures are described vaguely. Rules make the decision policy inspectable and changeable by local clinicians, but they do not remove the need for local clinical review and fairness testing on real records before any operational use.

VII. CONCLUSION

Seizure frequency is a core outcome measure in epilepsy, recorded almost exclusively in narrative clinic letters. This paper proposes a training-free pipeline with two stages: one LLM call extracts every candidate fact with its quoted evidence in a canonical form, then a decision stage applies a fixed policy to those candidates.

Both Hybrid (Purist F1: 0.86) and LLM-only (0.85) exceeded the benchmark previously reported for a fine-tuned LLM (0.81) on a different held-out sample of the same synthetic corpus. The extraction call's own provisional answer reaches 0.79; applying the decision policy to the evidence it found adds a further +0.06 to +0.07, whether that policy is applied by recorded rules or by a second LLM call. The result depends on the extraction prompt's examples, closed label forms and evidence obligation, and it held across six models, including two run on a single laptop GPU.

Automatically extracting seizure frequency patterns from EHRs could make this outcome measure available for large-scale clinical research: cohort identification, retrospective studies and predictive modelling. This method reaches the reported benchmark without any data or compute for fine-tuning. Future research should evaluate it on real patient data and on a broader set of clinical information extraction tasks before any of those uses.

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